MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.3 Worksheet

1. Which of the following expressions are defined? Which are meaningless? Explain.

a) $(\mathbf{a}\cdot\mathbf{b})\cdot\mathbf{c}$

This expression is meaningless since $(\mathbf{a}\cdot\mathbf{b})$ represents a scalar and you can't take the dot product of a scalar and the vector **c**.

b) $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

This expression is defined. $(\mathbf{a}\cdot\mathbf{b})$ represents a scalar and then this scalar is being multiplied to the vector **c**.

c) $\|\mathbf{a}\|(\mathbf{b}\cdot\mathbf{c})$

This expression is defined. $(\mathbf{b}\cdot\mathbf{c})$ represents a scalar and then this scalar is being multiplied to the scalar $\|\mathbf{a}\|$.

d) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$

This expression is defined. $(\mathbf{b}+\mathbf{c})$ represents a vector and this vector is then being dotted with the vector **a**.

e) $\mathbf{a}\cdot\mathbf{b}+\mathbf{c}$

This expression is meaningless. Since there are no parentheses we deal with each operation from left to right. $\mathbf{a}\cdot\mathbf{b}$ results in a scalar. This is then attempting to be added to the vector **c** which doesn't make sense.

f) $\|\mathbf{a}\|\cdot(\mathbf{b}+\mathbf{c})$

This expression is meaningless. Since $\|\mathbf{a}\|$ represents a scalar, it doesn't make sense to use this in a dot product calculation.

2. Find $\mathbf{a}\cdot\mathbf{b}$ for each of the following sets of vectors and determine the angle between each of these vectors. Round angles to 2 decimal places and express these angles in radians.

a) $\mathbf{a}=\langle -2,3\rangle$ & $\mathbf{b}=\langle 0.7,1.2\rangle$

$$\mathbf{a}\cdot\mathbf{b}=(-2)(0.7)+(3)(1.2)=-1.4+3.6=2.2$$

$$\cos(\theta)=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} \rightarrow \cos(\theta)=\frac{2.2}{\sqrt{13}\cdot\sqrt{1.93}} \rightarrow \theta=\cos^{-1}\left(\frac{2.2}{\sqrt{13}\cdot\sqrt{1.93}}\right)\approx 1.12$$

b) $\mathbf{a}=\langle 6,-2,3\rangle$ & $\mathbf{b}=\langle 2,5,-1\rangle$

$$\mathbf{a}\cdot\mathbf{b}=(6)(2)+(-2)(5)+(3)(-1)=-1$$

$$\cos(\theta)=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}\ \rightarrow\ \cos(\theta)=\frac{-1}{\sqrt{49}\cdot\sqrt{30}}\ \rightarrow\ \theta=\cos^{-1}\left(\frac{-1}{7\sqrt{30}}\right)\approx 1.60$$

c) $\mathbf{a}=2\mathbf{i}+\mathbf{j}$ & $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$

$$\mathbf{a}\cdot\mathbf{b}=(2)(1)+(1)(-1)+(0)(1)=1$$

$$\cos(\theta)=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}\ \rightarrow\ \cos(\theta)=\frac{1}{\sqrt{5}\cdot\sqrt{3}}\ \rightarrow\ \theta=\cos^{-1}\left(\frac{1}{\sqrt{15}}\right)\approx 1.31$$

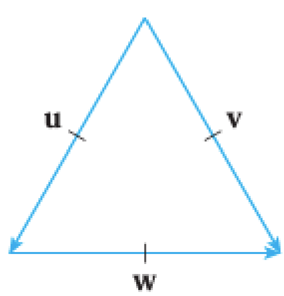
d) $\mathbf{a}=4\mathbf{i}-3\mathbf{j}+\mathbf{k}$ & $\mathbf{b}=2\mathbf{i}-\mathbf{k}$

$$\mathbf{a}\cdot\mathbf{b}=(4)(2)+(-3)(0)+(1)(-1)=7$$

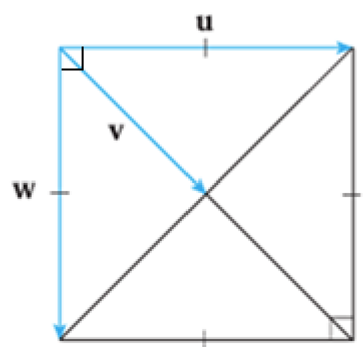
$$\cos(\theta)=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}\ \rightarrow\ \cos(\theta)=\frac{7}{\sqrt{26}\cdot\sqrt{5}}\ \rightarrow\ \theta=\cos^{-1}\left(\frac{7}{\sqrt{26}\cdot\sqrt{5}}\right)\approx 0.91$$

3. If $\mathbf{u}$ is a unit vector, find $\mathbf{u}\cdot\mathbf{v}$ and $\mathbf{u}\cdot\mathbf{w}$.

a)



b)



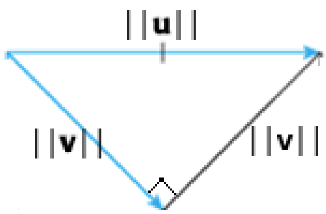
a) We see that all vectors here are the same length and therefore the triangle here is an equilateral triangle which also implies that the angle between all vectors is $60°=\frac{\pi}{3}$.

Therefore,

$$\mathbf{u}\cdot\mathbf{v}=\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta)=1\cdot1\cdot\cos\left(\frac{\pi}{3}\right)=\frac{1}{2}$$

$$\mathbf{u}\cdot\mathbf{w}=\|\mathbf{u}\|\|\mathbf{w}\|\cos(\theta)=1\cdot1\cdot\cos\left(\frac{\pi}{3}\right)=\frac{1}{2}$$

b) We the angle between vectors **u** and **w** is $90° = \frac{\pi}{2}$. Since **v** represents a portion of the diagonal of the given square this implies that the angle between **u** and **v** is $45° = \frac{\pi}{4}$. We also note that the diagonals of a square intersect at a right angle and so the triangle with side lengths of **u** and **v** is a right triangle (see the picture below). Therefore, by the Pythagorean Theorem we have that $\|\mathbf{v}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 \;\rightarrow\; 2\|\mathbf{v}\|^2 = 1 \;\rightarrow\; \|\mathbf{v}\| = \frac{1}{\sqrt{2}}$.



Therefore,

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) = 1\cdot\frac{1}{\sqrt{2}}\cdot\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{2}}{2} = \frac{1}{2}$$

$$\mathbf{u}\cdot\mathbf{w} = \|\mathbf{u}\|\|\mathbf{w}\|\cos(\theta) = 1\cdot 1\cdot\cos\left(\frac{\pi}{2}\right) = 0$$

Note we could have also concluded that since **u** and **w** are perpendicular, we know that $\mathbf{u}\cdot\mathbf{w} = 0$.

4. Determine the angle between the vector $\mathbf{u} = \langle 1,2,3\rangle$ and each of the coordinate axes. Round angles to 2 decimal places and express these angles in radians.

To find the angle between **u** and the *x*-axis we will find the angle between the vectors **u** and $\mathbf{i} = \langle 1,0,0\rangle$.

$$\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{i}}{\|\mathbf{u}\|\|\mathbf{i}\|} \;\rightarrow\; \cos(\theta) = \frac{1}{\sqrt{14}} \;\rightarrow\; \theta = \cos^{-1}\left(\frac{1}{\sqrt{14}}\right) \approx 1.30$$

To find the angle between **u** and the *y*-axis we will find the angle between the vectors **u** and $\mathbf{j} = \langle 0,1,0\rangle$.

$$\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{j}}{\|\mathbf{u}\|\|\mathbf{j}\|} \;\rightarrow\; \cos(\theta) = \frac{2}{\sqrt{14}} \;\rightarrow\; \theta = \cos^{-1}\left(\frac{2}{\sqrt{14}}\right) \approx 1.01$$

To find the angle between **u** and the *z*-axis we will find the angle between the vectors **u** and $\mathbf{k} = \langle 0,0,1\rangle$.

$$\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{k}}{\|\mathbf{u}\|\|\mathbf{k}\|} \;\rightarrow\; \cos(\theta) = \frac{3}{\sqrt{14}} \;\rightarrow\; \theta = \cos^{-1}\left(\frac{3}{\sqrt{14}}\right) \approx 0.64$$

5. Determine if the given vectors are orthogonal, parallel, or neither.

a) $\mathbf{a} = \langle -5, 3, 7 \rangle$ & $\mathbf{b} = \langle 6, -8, 2 \rangle$

We check that $\mathbf{a} \cdot \mathbf{b} = (-5)(6) + (3)(-8) + (7)(2) = -30 - 24 + 14 = -40$. This means that the vectors are not orthogonal.

We can also see that these vectors are not scalar multiples of each other, so they are not parallel.

Therefore, these vectors are neither orthogonal nor parallel.

b) $\mathbf{a} = \langle 4, 6 \rangle$ & $\mathbf{b} = \langle -3, 2 \rangle$

We check that $\mathbf{a} \cdot \mathbf{b} = (4)(-3) + (6)(2) = -12 + 12 = 0$. This means that the vectors are orthogonal.

c) $\mathbf{a} = -\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ & $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$

We check that $\mathbf{a} \cdot \mathbf{b} = (-1)(3) + (2)(4) + (5)(-1) = -3 + 8 - 5 = 0$. This means that the vectors are orthogonal.

d) $\mathbf{a} = 2\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}$ & $\mathbf{b} = -3\mathbf{i} - 9\mathbf{j} + 6\mathbf{k}$

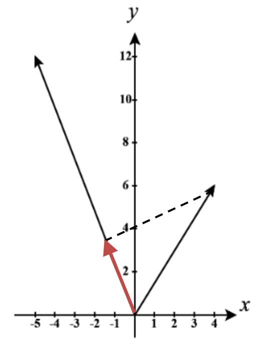
We check that $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (6)(-9) + (-4)(6) = -6 - 54 - 24 = -84$. This means that the vectors are not orthogonal.

We see that these vectors are scalar multiplies of each other since $-\frac{3}{2}\mathbf{a} = \mathbf{b}$.

Therefore, these vectors are parallel.

6. Find the component and projection of **b** onto **a**. In the 2-dimensional cases include a sketch of these vectors and the projection (sketch all vectors starting from the origin).

a) $\mathbf{a}=\langle -5,12\rangle$ & $\mathbf{b}=\langle 4,6\rangle$

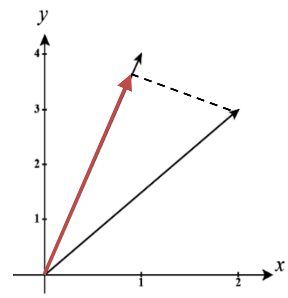


$$comp_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}=\frac{(-5)(4)+(12)(6)}{\sqrt{169}}=\frac{52}{13}=4$$

$$proj_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}=\frac{52}{169}\langle -5,12\rangle=\left\langle -\frac{20}{13},\frac{48}{13}\right\rangle$$

This projection vector is shown in red.

b) $\mathbf{a}=\langle 1,4\rangle$ & $\mathbf{b}=\langle 2,3\rangle$



$$comp_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}=\frac{(1)(2)+(4)(3)}{\sqrt{17}}=\frac{14}{\sqrt{17}}$$

$$proj_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}=\frac{14}{17}\langle 1,4\rangle=\left\langle \frac{14}{17},\frac{56}{17}\right\rangle$$

This projection vector is shown in red.

c) $\mathbf{a}=\langle -2,3.-6\rangle$ & $\mathbf{b}=\langle 5,-1,4\rangle$

$$comp_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}=\frac{(-2)(5)+(3)(-1)+(-6)(4)}{\sqrt{49}}=\frac{-37}{7}$$

$$proj_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}=\frac{-37}{49}\langle -2,3,-6\rangle=\left\langle \frac{74}{49},-\frac{111}{49},\frac{222}{49}\right\rangle$$

d) $\mathbf{a}=2\mathbf{i}-\mathbf{j}+4\mathbf{k}$ & $\mathbf{b}=\mathbf{j}+\frac{1}{2}\mathbf{k}$

$$comp_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}=\frac{(2)(0)+(-1)(1)+(4)\left(\frac{1}{2}\right)}{\sqrt{21}}=\frac{1}{\sqrt{21}}$$

$$proj_{\mathbf{a}}(\mathbf{b})=\frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}=\frac{1}{21}\langle 2,-1,4\rangle=\left\langle \frac{2}{21},-\frac{1}{21},\frac{4}{21}\right\rangle$$

7. Suppose that **a** and **b** are nonzero vectors.

a) Under what circumstances is $comp_{\mathbf{a}}(\mathbf{b}) = comp_{\mathbf{b}}(\mathbf{a})$.

We know that $comp_{\mathbf{a}}(\mathbf{b}) = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}$ & $comp_{\mathbf{b}}(\mathbf{a}) = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|}$. We can then see that $comp_{\mathbf{a}}(\mathbf{b}) = comp_{\mathbf{b}}(\mathbf{a})$ whenever $\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|} = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|}$. This occurs if $\mathbf{a}\cdot\mathbf{b} = 0$ or if $\|\mathbf{a}\| = \|\mathbf{b}\|$. That is, if **a** and **b** are orthogonal or if they have the same length.

b) Under what circumstances is $proj_{\mathbf{a}}(\mathbf{b}) = proj_{\mathbf{b}}(\mathbf{a})$.

We know that $proj_{\mathbf{a}}(\mathbf{b}) = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}$ & $proj_{\mathbf{b}}(\mathbf{a}) = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}$. We can then see that $proj_{\mathbf{a}}(\mathbf{b}) = proj_{\mathbf{b}}(\mathbf{a})$ whenever $\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a} = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}$. This can only happen if $\mathbf{a}\cdot\mathbf{b} = 0$ or $\dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|^2}$.

In order for $\dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|^2}$, these vectors must have the same magnitude and therefore

$\dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|^2} \;\rightarrow\; \dfrac{\|\mathbf{a}\|}{\|\mathbf{a}\|^2} = \dfrac{\|\mathbf{b}\|}{\|\mathbf{b}\|^2} \;\rightarrow\; \dfrac{1}{\|\mathbf{a}\|} = \dfrac{1}{\|\mathbf{b}\|} \;\rightarrow\; \|\mathbf{a}\| = \|\mathbf{b}\|$. However, if we substitute this into $\dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|^2}$ we get that $\dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|^2} \;\rightarrow\; \dfrac{\mathbf{a}}{\|\mathbf{a}\|^2} = \dfrac{\mathbf{b}}{\|\mathbf{a}\|^2} \;\rightarrow\; \mathbf{a} = \mathbf{b}$

Therefore, $proj_{\mathbf{a}}(\mathbf{b}) = proj_{\mathbf{b}}(\mathbf{a})$ whenever $\mathbf{a}\cdot\mathbf{b} = 0$ or $\mathbf{a} = \mathbf{b}$. That is, when **a** and **b** are orthogonal or equal.

8. If $\mathbf{c} = \|\mathbf{a}\|\mathbf{b} + \|\mathbf{b}\|\mathbf{a}$, where **a**, **b**, and **c** are all nonzero vectors, show that **c** bisects the angle between **a** and **b.**

If **c** bisects the angle between **a** and **b**, then this means that the angle between **a&c** should equal the angle between **b&c**.

We first note that based on the definition of **c**

$$\frac{\mathbf{a}\cdot\mathbf{c}}{\|\mathbf{a}\|\|\mathbf{c}\|} = \frac{\|\mathbf{a}\|\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|\mathbf{a}\cdot\mathbf{a}}{\|\mathbf{a}\|\|\mathbf{c}\|} = \frac{\|\mathbf{a}\|\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|\|\mathbf{a}\|^2}{\|\mathbf{a}\|\|\mathbf{c}\|} = \frac{\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|\|\mathbf{a}\|}{\|\mathbf{c}\|} \text{ and}$$

$$\frac{\mathbf{b}\cdot\mathbf{c}}{\|\mathbf{b}\|\|\mathbf{c}\|} = \frac{\|\mathbf{a}\|\mathbf{b}\cdot\mathbf{b} + \|\mathbf{b}\|\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|\|\mathbf{c}\|} = \frac{\|\mathbf{a}\|\|\mathbf{b}\|^2 + \|\mathbf{b}\|\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|\|\mathbf{c}\|} = \frac{\|\mathbf{b}\|\|\mathbf{a}\| + \mathbf{a}\cdot\mathbf{b}}{\|\mathbf{c}\|}$$

Therefore, we calculate that the angle between **a&c** is $\theta_1 = \cos^{-1}\left(\frac{\mathbf{a}\cdot\mathbf{c}}{\|\mathbf{a}\|\|\mathbf{c}\|}\right) = \cos^{-1}\left(\frac{\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|\|\mathbf{a}\|}{\|\mathbf{c}\|}\right)$ and the angle between **b&c** is $\theta_2 = \cos^{-1}\left(\frac{\mathbf{b}\cdot\mathbf{c}}{\|\mathbf{b}\|\|\mathbf{c}\|}\right) = \cos^{-1}\left(\frac{\|\mathbf{b}\|\|\mathbf{a}\| + \mathbf{a}\cdot\mathbf{b}}{\|\mathbf{c}\|}\right)$.

So clearly $\theta_1 = \theta_2$.

9. Show that if $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal, then the vectors **u** and **v** must have the same length.

If these vectors are orthogonal, this means that

$$\begin{aligned}
(\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = 0 \quad &\to \quad \mathbf{u}\cdot\mathbf{u} + (\mathbf{u}\cdot\mathbf{v}) - (\mathbf{u}\cdot\mathbf{v}) - \mathbf{v}\cdot\mathbf{v} = 0 \\
&\to \quad \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 = 0 \\
&\to \quad \|\mathbf{u}\|^2 = \|\mathbf{v}\|^2 \\
&\to \quad \|\mathbf{u}\| = \|\mathbf{v}\|
\end{aligned}$$

Therefore, **u** and **v** must have the same length.

10. If $\mathbf{r}=\langle x,y,z\rangle$, $\mathbf{a}=\langle a_1,a_2,a_3\rangle$, & $\mathbf{b}=\langle b_1,b_2,b_3\rangle$, show that the vector equation $(\mathbf{r}-\mathbf{a})\cdot(\mathbf{r}-\mathbf{b})=0$ represents a sphere, and finds its center and radius.

$$(\mathbf{r}-\mathbf{a})\cdot(\mathbf{r}-\mathbf{b})=0 \quad\rightarrow\quad \left(\langle x,y,z\rangle-\langle a_1,a_2,a_3\rangle\right)\cdot\left(\langle x,y,z\rangle-\langle b_1,b_2,b_3\rangle\right)=0$$

$$\rightarrow\quad \langle x-a_1,y-a_2,z-a_3\rangle\cdot\langle x-b_1,y-b_2,z-b_3\rangle=0$$

$$\rightarrow\quad (x-a_1)(x-b_1)+(y-a_2)(y-b_2)+(z-a_3)(z-b_3)=0$$

$$\rightarrow\quad \left(x^2-(a_1+b_1)x+a_1b_1\right)+\left(y^2-(a_2+b_2)y+a_2b_2\right)+\left(z^2-(a_3+b_3)z+a_3b_3\right)=0$$

$$\rightarrow\quad \left(x^2-(a_1+b_1)x+\_\_\_\_\right)+\left(y^2-(a_2+b_2)y+\_\_\_\_\right)+\left(z^2-(a_3+b_3)z+\_\_\_\_\right)=-a_1b_1-a_2b_2-a_3b_3$$

$$\rightarrow\quad \left(x^2-(a_1+b_1)x+\left(\frac{a_1+b_1}{2}\right)^2\right)+\left(y^2-(a_2+b_2)y+\left(\frac{a_2+b_2}{2}\right)^2\right)+\left(z^2-(a_3+b_3)z+\left(\frac{a_3+b_3}{2}\right)^2\right)$$
$$=-a_1b_1-a_2b_2-a_3b_3+\left(\frac{a_1+b_1}{2}\right)^2+\left(\frac{a_2+b_2}{2}\right)^2+\left(\frac{a_3+b_3}{2}\right)^2$$

$$\rightarrow\quad \left(x-\frac{a_1+b_1}{2}\right)^2+\left(y-\frac{a_2+b_2}{2}\right)^2+\left(z-\frac{a_3+b_3}{2}\right)^2$$
$$=-a_1b_1-a_2b_2-a_3b_3+\frac{1}{4}\left(a_1^2+2a_1b_1+b_1^2+a_2^2+2a_2b_2+b_2^2+a_3^2+2a_3b_3+b_3^2\right)$$

$$\rightarrow\quad \left(x-\frac{a_1+b_1}{2}\right)^2+\left(y-\frac{a_2+b_2}{2}\right)^2+\left(z-\frac{a_3+b_3}{2}\right)^2$$
$$=\frac{1}{4}\left(a_1^2-2a_1b_1+b_1^2+a_2^2-2a_2b_2+b_2^2+a_3^2-2a_3b_3+b_3^2\right)$$

$$\rightarrow\quad \left(x-\frac{a_1+b_1}{2}\right)^2+\left(y-\frac{a_2+b_2}{2}\right)^2+\left(z-\frac{a_3+b_3}{2}\right)^2=\frac{1}{4}\left((a_1-b_1)^2+(a_2-b_2)^2+(a_3-b_3)^2\right)$$

This means we have a sphere with center $\left(\frac{a_1+b_1}{2},\frac{a_2+b_2}{2},\frac{a_3+b_3}{2}\right)$ and radius

$$r=\frac{1}{2}\sqrt{(a_1-b_1)^2+(a_2-b_2)^2+(a_3-b_3)^2}$$